ECHO ESSENTIALS | REVISION KIT

POCKET CARDS AND 24 MCQs

Rapid thresholds. Exam-style questions. Full explanations.

4 print-ready pocket cards (AS, low-flow AS, regurgitant valves, function and right heart)
24 single-best-answer questions with guideline-based explanations
Answer key and score sheet

Dr. [Your Name]
Consultant Cardiologist & Echocardiographer
Edition 1 | October 2026

POCKET CARD 1

Aortic stenosis grading

Gradient, area, flow, EF.

Severity

	Moderate	Severe	Very severe / critical
Vmax (m/s)	3.0 - 3.9	≥ 4.0	≥ 5.0
MG (mmHg)	20 - 39	≥ 40	≥ 60
AVA (cm²)	1.0 - 1.5	≤ 1.0	typically < 0.6
AVAi (cm²/m²)	0.60 - 0.85	≤ 0.6	
DVI	0.25 - 0.50	≤ 0.25	

Equations

AVA = LVOT area × VTI_{LVOT} / VTI_{AV} | DVI = VTI_{LVOT} / VTI_{AV} | SVi = LVOT area × VTI_{LVOT} / BSA | Low flow: SVi < 35 mL/m²

Critical AS

Adult: Vmax ≥ 5 m/s and/or MG ≥ 60 mmHg, usually AVA < 0.6 cm² (ACC/AHA: very severe). No single universal definition: report the numbers. Neonatal critical AS is a different, duct-dependent entity.

Before you call it

Multiple windows and a dedicated probe. Alignment errors underestimate. BP controlled. LVOT area: consider 3D or CT. Average 5 beats in AF.

POCKET CARD 2

Low-flow, low-gradient AS

Two types, one question: is it truly severe?

The two types

	Reduced EF (stage D2)	Preserved EF - paradoxical (stage D3)
Definition	AVA ≤ 1.0, MG < 40, LVEF < 50%, SVi < 35	AVA ≤ 1.0 (AVAi ≤ 0.6), MG < 40, LVEF ≥ 50%, SVi < 35
Patient	Dilated LV, cardiomyopathy	Older, women, small stiff LV, hypertension
Test	Low-dose DSE and/or CT calcium	Exclude errors, then CT calcium
Trap	Pseudo-severe AS	Measurement error (BP, LVOT area, BSA)

DSE

Dobutamine 5 to 20 µg/kg/min. Flow reserve = SV rise ≥ 20%. True-severe: AVA stays ≤ 1.0 with Vmax ≥ 4 m/s. Pseudo-severe: AVA rises above 1.0-1.2. No flow reserve: use CT calcium; AVR still beneficial.

CT calcium (Agatston)

	Men	Women
Severe very likely	≥ 3000	≥ 1600
Severe likely	≥ 2000	≥ 1200
Severe unlikely	< 1600	< 800

Normal-flow low-gradient

AVA ≤ 1.0, MG < 40, SVi ≥ 35, LVEF ≥ 50%: usually moderate AS or measurement inconsistency.

POCKET CARD 3

Regurgitant valves

Severe thresholds at a glance.

Severe criteria

	AR	MR (primary)	TR
Vena contracta	> 6 mm	≥ 7 mm	≥ 7 mm
EROA	≥ 0.30 cm²	≥ 0.40 cm²	≥ 40 mm²
RVol	≥ 60 mL	≥ 60 mL	≥ 45 mL
RF	≥ 50%	≥ 50%	-
Other	PHT < 200 ms; holodiastolic reversal; jet/LVOT ≥ 65%	PV systolic reversal; E > 1.5 m/s; PISA ≥ 10 mm	Hepatic systolic reversal; PISA > 9 mm; dense triangular CW

TR beyond severe

Massive: VC 14-20 mm, EROA 60-79 mm², RVol 60-74 mL. Torrential: VC ≥ 21 mm, EROA ≥ 80 mm², RVol ≥ 75 mL.

Secondary MR

ESC/ACC severe: EROA ≥ 0.20 cm² or RVol ≥ 30 mL (elliptical orifice). COAPT enrolled EROA > 0.30.

Mitral stenosis

Clinically significant MVA ≤ 1.5 cm²; PHT ≥ 150 ms; very severe MVA ≤ 1.0, PHT ≥ 220. MVA = 220/PHT. Wilkins ≤ 8 favours PBC. Exclude LA thrombus and more than mild MR.

Triggers for surgery

AR: symptoms; LVEF ≤ 50-55%; LVESD > 50 mm; LVEDD > 70 mm. **Primary MR**: symptoms; LVESD ≥ 40 mm; LVEF ≤ 60%; new AF; PASP > 50; LAVI ≥ 60.

POCKET CARD 4

Function, diastology, right heart

The numbers you quote every day.

LV and LA

	Men	Women
LVEF	52 - 72%	54 - 74%
LVEDD	42 - 58 mm	38 - 52 mm
LVMI (2D)	50 - 102	44 - 88 g/m²
LAVI	16 - 34	16 - 34 mL/m²
RWT	≤ 0.42	≤ 0.42

Raised LAP markers

Septal e' < 7, lateral e' < 10, average E/e' > 14, TR > 2.8 m/s, LAVI > 34. AF: septal E/e' ≥ 11, DT ≤ 160 ms. Use the ASE 2025 algorithm for grading and HFpEF diagnosis.

RV

TAPSE < 17 mm | S' < 9.5 cm/s | FAC < 35% | FWLS less negative than -20% | basal diameter > 41 mm | RA area > 18 cm².

PH

PASP = 4V² + RAP (3 / 8 / 15). TR ≤ 2.8 low; 2.9-3.4 intermediate; > 3.4 high (with other PH signs raising one level).

Dr. [Your Name] | Licensed to single user. Do not redistribute.

PART TWO

24 MCQs: single best answer

Mark your answers on the score sheet at the end. Answers and explanations follow on the next pages.

1. A patient has Vmax 4.6 m/s, mean gradient 52 mmHg, AVA 0.8 cm², SVi 42 mL/m². Which best describes the stenosis?

A. Moderate AS
B. Severe high-gradient AS
C. Paradoxical low-flow AS
D. Pseudo-severe AS

2. Which values define very severe (critical) adult aortic stenosis?

A. Vmax ≥ 4 m/s and MG ≥ 40
B. Vmax ≥ 5 m/s and/or MG ≥ 60 mmHg
C. AVA < 1.5 cm²
D. DVI < 0.5

3. Low flow in aortic stenosis is defined as:

A. SVi < 35 mL/m²
B. LVEF < 50%
C. MG < 40 mmHg
D. AVA < 1.5 cm²

4. A 66-year-old man: LVEF 28%, AVA 0.8 cm², MG 27 mmHg, SVi 26 mL/m². Best next test?

A. Repeat TTE in 6 months
B. Low-dose dobutamine stress echo and/or CT calcium score
C. Exercise stress test
D. Cardiac MRI for LGE

5. During DSE the stroke volume increases by 25%, AVA rises to 1.4 cm² and Vmax remains 3.2 m/s. Interpretation?

A. True-severe AS
B. Pseudo-severe AS with flow reserve
C. Non-diagnostic, no flow reserve
D. Paradoxical LFLG

6. Which is TRUE of paradoxical low-flow, low-gradient AS?

A. LVEF < 50%
B. SVi < 35 with LVEF ≥ 50%
C. Dobutamine is the preferred test
D. It always represents moderate AS

7. CT aortic valve calcium score making severe AS very likely in a woman:

A. ≥ 800 AU
B. ≥ 1200 AU
C. ≥ 1600 AU
D. ≥ 3000 AU

8. What is the effect of underestimating the LVOT diameter by 1 mm?

A. No change
B. AVA overestimated
C. AVA underestimated, since the diameter is squared
D. Gradient underestimated

9. DVI in severe AS is:

A. ≤ 0.25
B. 0.25-0.50
C. > 0.5
D. > 1.0

10. Best way to improve Vmax acquisition in AS:

A. Use the imaging probe from the apex only
B. Multiple windows with a dedicated non-imaging probe
C. Increase gain
D. Average gradients from three beats

11. Severe AR is suggested by all EXCEPT:

A. Vena contracta > 6 mm
B. PHT < 200 ms
C. Holodiastolic flow reversal in the abdominal aorta
D. Jet width / LVOT < 25%

12. Chronic severe AR, asymptomatic, LVEF 58%, LVESD 52 mm. Management?

A. Surveillance
B. Surgery should be considered
C. Medical therapy only
D. Stress echo

13. Mitral stenosis: which measurement is unreliable after balloon valvotomy and with significant AR?

A. 2D planimetry
B. Pressure half-time
C. Mean gradient
D. PISA

14. A Wilkins score of 7 in a symptomatic patient with MVA 1.1 cm² and no LA thrombus suggests:

A. Mitral valve replacement
B. Percutaneous balloon commissurotomy
C. Medical therapy alone
D. Observation

15. Primary MR: vena contracta 8 mm, EROA 0.45 cm², RVol 65 mL, LVESD 41 mm, no symptoms. Best action?

A. Surveillance in 12 months
B. Surgery (repair if feasible)
C. TEER
D. Start ACE inhibitor only

16. Wall-hugging MR jets are best assessed by:

A. Colour jet area
B. Vena contracta, PV flow, PISA and quantitative methods
C. Mitral E velocity only
D. LA size

17. Secondary MR is classed as severe at which EROA (ESC/ACC practice)?

A. ≥ 0.10
B. ≥ 0.20
C. ≥ 0.40 only
D. ≥ 0.60

18. TR with vena contracta 15 mm, EROA 65 mm². Grade?

A. Severe
B. Massive
C. Torrential
D. Moderate

19. TR peak velocity 2.5 m/s but no other PH signs. PH probability?

A. High
B. Intermediate
C. Low
D. Cannot be assessed

20. Estimated RAP with IVC 24 mm and < 50% collapse:

A. 3
B. 8
C. 15
D. 20

21. Which parameter indicates RV systolic dysfunction?

A. TAPSE 20 mm
B. S' 12 cm/s
C. FAC 30%
D. RA area 16 cm²

22. Which is NOT a marker of raised LAP in the ASE diastology approach?

A. Average E/e' > 14
B. TR velocity > 2.8 m/s
C. LAVI > 34 mL/m² alone, in an athlete
D. Septal e' < 7 cm/s

23. Which GLS change suggests cardiotoxicity?

A. Relative fall > 15% from baseline
B. Absolute value of -20%
C. Fall of 2%
D. LAVI > 34

24. The first sentence of an echo report should:

A. List every measurement
B. Answer the clinical question
C. Describe image quality
D. Recommend therapy

ANSWER KEY

Answers and explanations

1. B - Severe high-gradient AS

Vmax ≥ 4 m/s and MG ≥ 40 mmHg with AVA ≤ 1.0 define severe high-gradient AS. Flow is normal (SVi ≥ 35). Not very severe, because Vmax < 5 and MG < 60.

2. B - Vmax ≥ 5 m/s and/or MG ≥ 60 mmHg

ACC/AHA calls Vmax ≥ 5 m/s or MG ≥ 60 mmHg very severe; ESC lists Vmax > 5 as a trigger for early intervention. There is no single universal definition of the word critical, so report the numbers.

3. A - SVi < 35 mL/m²

Low flow is SVi < 35 mL/m² and is independent of LVEF. LVEF defines which of the two low-flow types the patient has.

4. B - Low-dose dobutamine stress echo and/or CT calcium score

Classical low-flow, low-gradient AS with reduced EF. Distinguish true-severe from pseudo-severe by DSE and/or CT calcium. Exercise testing is not appropriate.

5. B - Pseudo-severe AS with flow reserve

Flow reserve is present (SV rise ≥ 20%), AVA exceeds 1.0-1.2 and gradients remain low: pseudo-severe (moderate) AS.

6. B - SVi < 35 with LVEF ≥ 50%

Paradoxical LFLG: LVEF ≥ 50%, SVi < 35 mL/m², AVA ≤ 1.0, MG < 40. Dobutamine cannot raise flow in the small stiff LV. Confirm severity with CT calcium after excluding errors.

7. C - ≥ 1600 AU

Women: ≥ 1600 very likely, ≥ 1200 likely, < 800 unlikely. Men: ≥ 3000, ≥ 2000, < 1600.

8. C - AVA underestimated, since the diameter is squared

LVOT area is π(d/2)², so small diameter errors are amplified. A smaller LVOT diameter gives a falsely small AVA and SVi. DVI avoids this error.

9. A - ≤ 0.25

DVI = VTI LVOT / VTI AV. Severe ≤ 0.25.

10. B - Multiple windows with a dedicated non-imaging probe

Use apical, right parasternal and suprasternal windows and report the highest velocity. Misalignment always underestimates.

11. D - Jet width / LVOT < 25%

Jet width / LVOT ≥ 65% is severe. < 25% is mild.

12. B - Surgery should be considered

LVESD > 50 mm is a threshold for surgery in asymptomatic severe AR. Always check the aortic root.

13. B - Pressure half-time

PHT depends on LA and LV compliance and is unreliable after PBC, with significant AR, AF, and tachycardia. Planimetry is preferred.

14. B - Percutaneous balloon commissurotomy

Wilkins ≤ 8 with favourable anatomy supports PBC, after excluding LA thrombus and more than mild MR.

15. B - Surgery (repair if feasible)

Severe primary MR with LVESD ≥ 40 mm: surgery is recommended; repair preferred.

16. B - Vena contracta, PV flow, PISA and quantitative methods

Coanda effect makes colour jet area underestimate eccentric jets.

17. B - ≥ 0.20

Lower thresholds apply because of the elliptical orifice (EROA ≥ 0.20 cm² or RVol ≥ 30 mL). COAPT enrolled > 0.30.

18. B - Massive

Massive: VC 14-20 mm, EROA 60-79 mm². Torrential: VC ≥ 21 mm, EROA ≥ 80 mm².

19. C - Low

TR ≤ 2.8 m/s with no other signs = low probability (ESC/ERS 2022).

20. C - 15

IVC > 21 mm and collapse < 50% = 15 mmHg.

21. C - FAC 30%

FAC < 35%. TAPSE < 17, S' < 9.5.

22. C - LAVI > 34 mL/m² alone, in an athlete

LAVI is a chronic marker and is enlarged in athletes, AF and other settings, so on its own it does not prove raised LAP. Interpret in context.

23. A - Relative fall > 15% from baseline

A relative drop of > 15% from baseline is clinically significant.

24. B - Answer the clinical question

The conclusion should answer the clinical question, with quantitative parameters supporting it.

Score: ____ / 24. Below 16: revisit Chapters 4 to 8. 16-21: solid. 22-24: exam-ready.

For educational use by qualified clinicians. Not a substitute for the full published guidelines, local protocols, or clinical judgement. Always verify thresholds against the current ASE, EACVI and ESC/EACTS documents.